20251007 rMZR in HII But No NGC4383

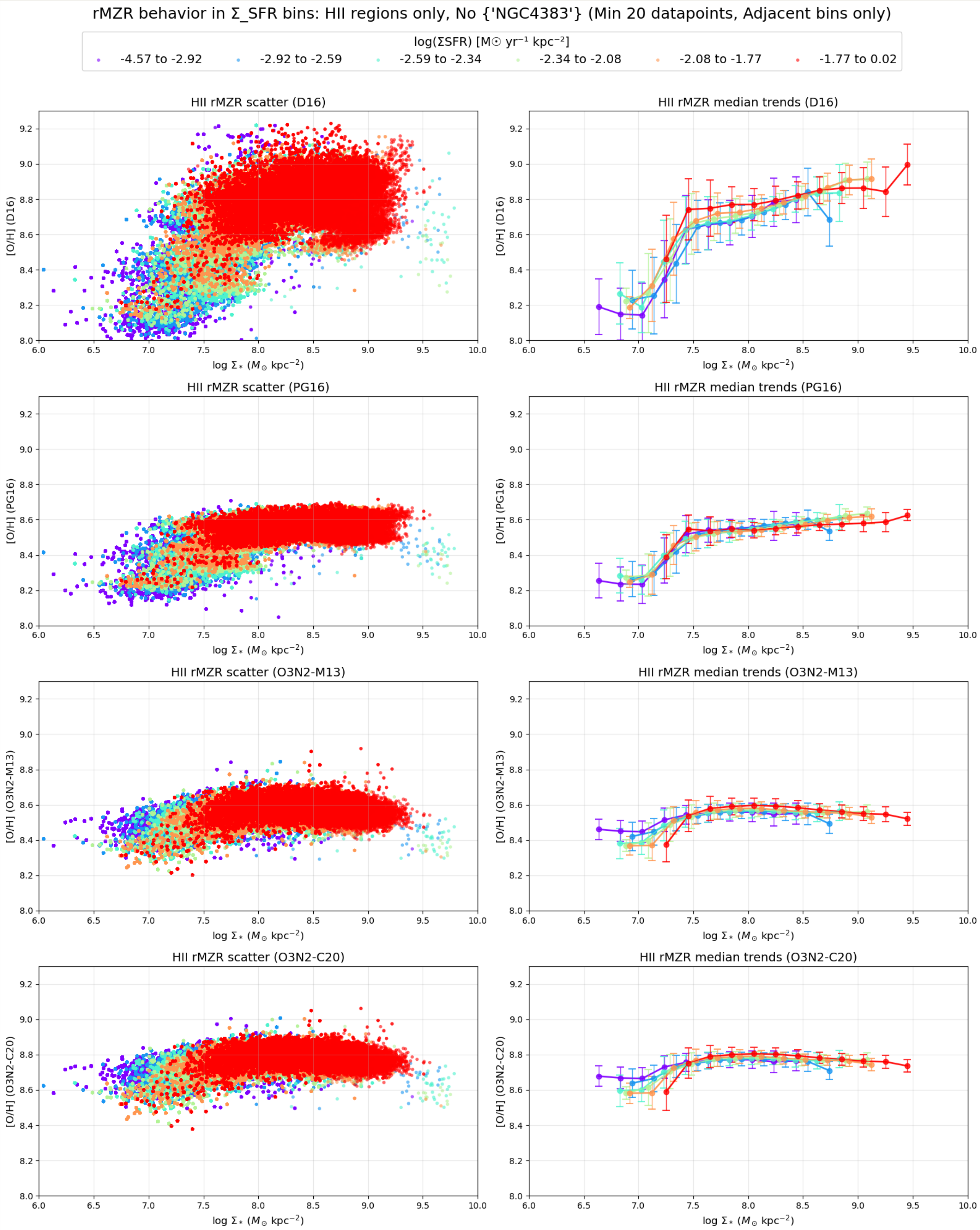
Note.

1. Try removing the abnnormal NGC4383. The reason is that it almost occupies the entire independent parameter space in rMZR. So if we want to study the SFR dependence in rMZR, it will be biased by this single galaxy. Therefore, we can try removing it and see how the results change.
2. Only spaxels from HII regions are included.
3. **Update to 6 $\Sigma_{\rm SFR}$ bins, which requires same spaxel counts in each $\Sigma_{\rm SFR}$ bin (so the binwidth will change in different panels).**
4. Stick with these 4 indicators: D16, PG16, O3N2-M13 and O3N2-C20.
5. Update to require at least 20 unique datapoints in each median bin of each $\Sigma_{\rm SFR}$ bin.

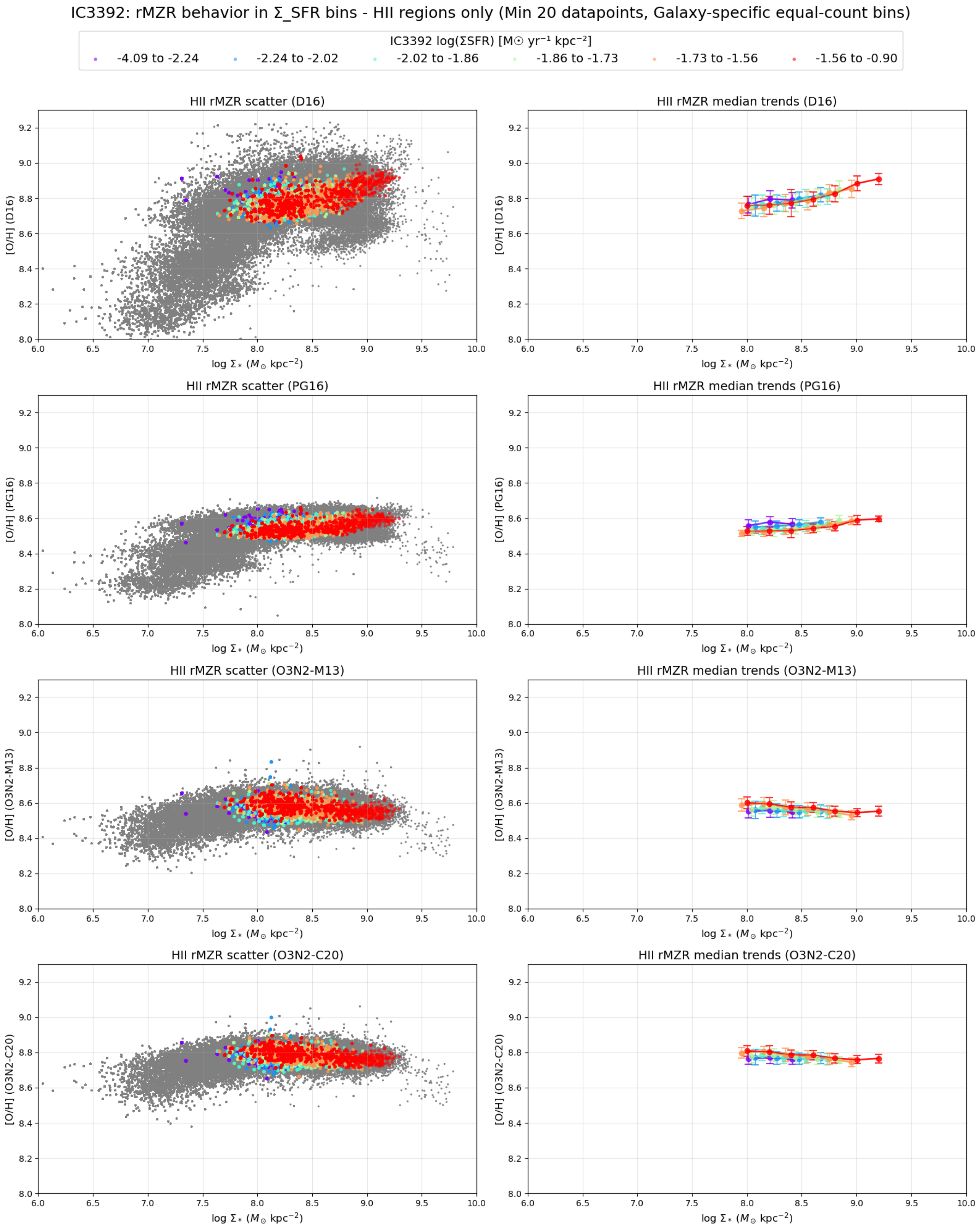
Copy table from previous notetable (* are those "abnormalies" in rMZR, together with the removed NGC4383):

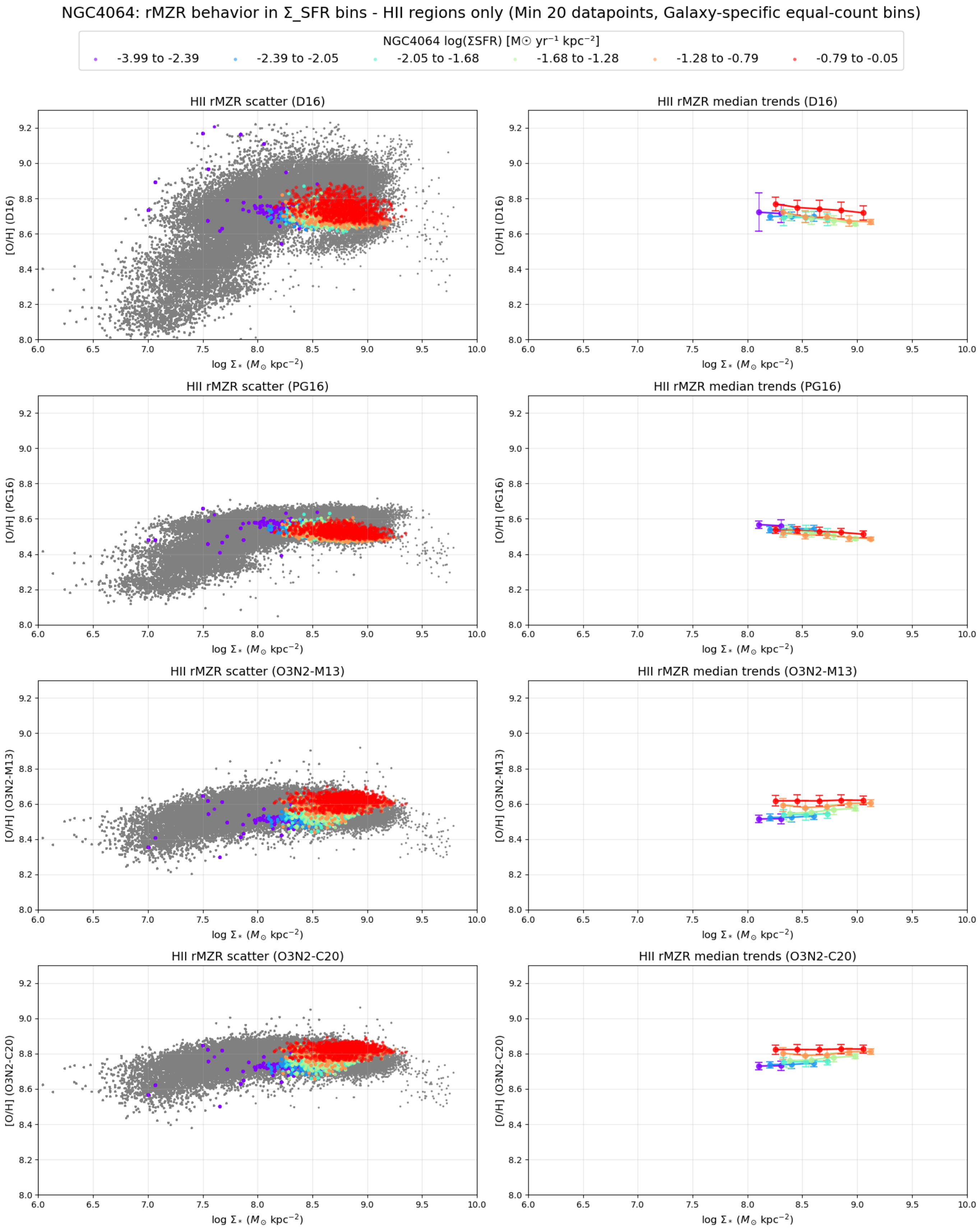
ID	POSITION	CORRELATION	EXPECTATION
IC3392	High	Positive	Yes
NGC4064	High	Positive	Yes
NGC4192	Cross	Inverse	Yes
NGC4293	Few (High?)	-	-
NGC4298	Cross	Positive (mostly?)	No?
NGC4330*	Low	Negative	Yes
NGC4396*	Low	Negative	Yes
NGC4419	High	Positive	Yes
NGC4457	Few (High?)	-	-
NGC4501	Cross	Positive	No
NGC4522*	Low	- (Positive?)	No?
NGC4694*	High	Positive	Yes
NGC4698	Few (Cross?)	-	-

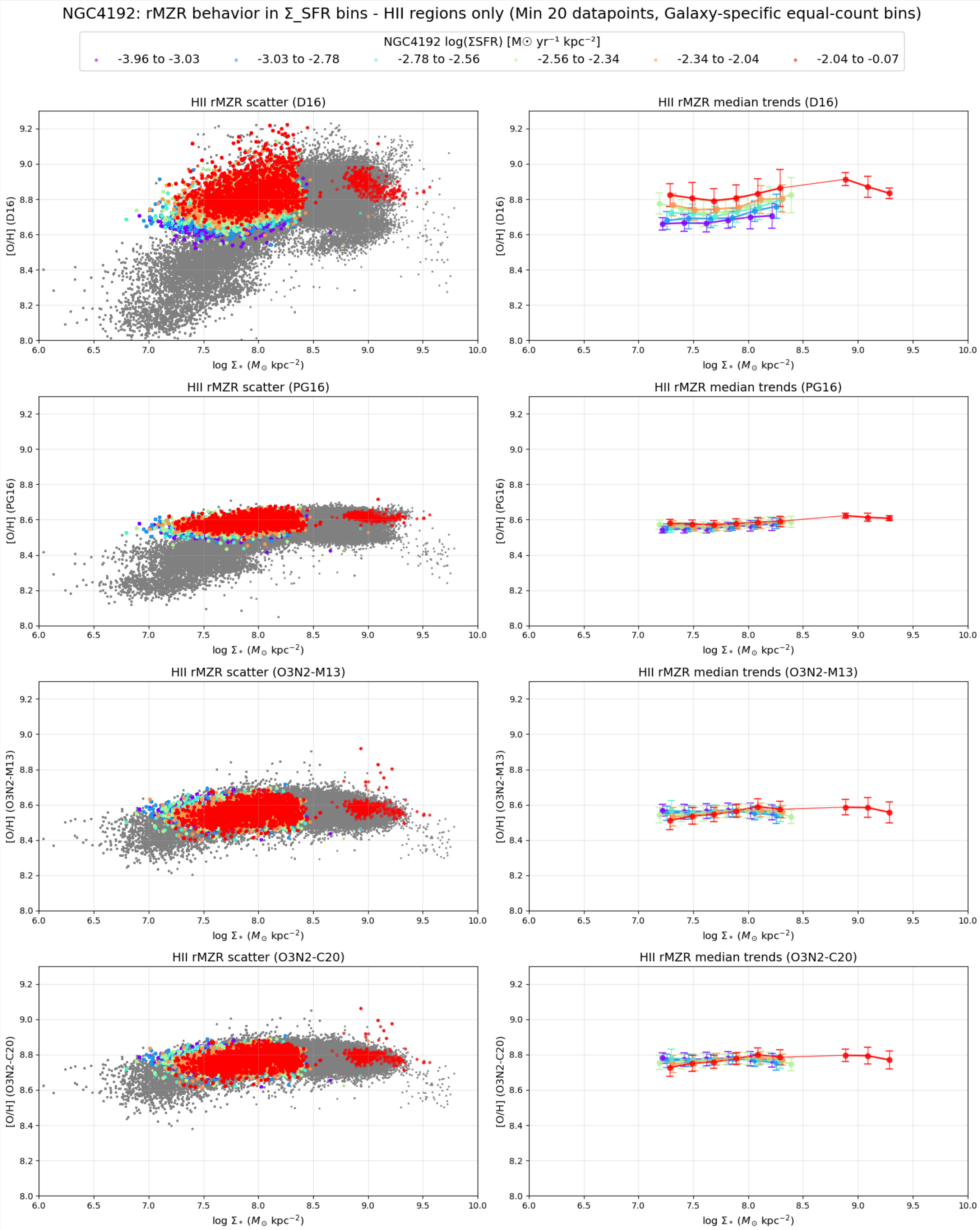
Combined

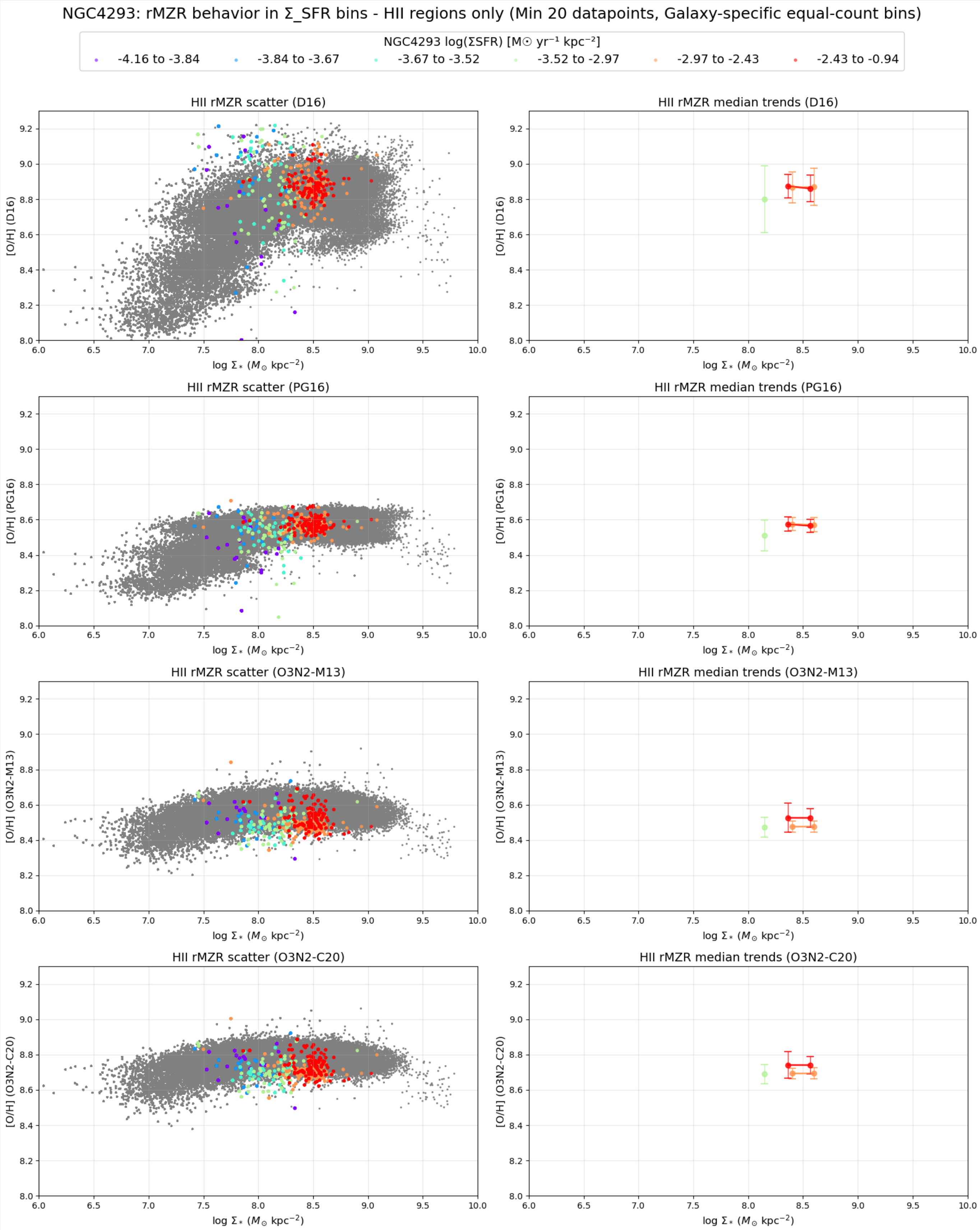

rMZR behavior in Σ_SFR bins: HII regions only, No {'NGC4383'} (Min 20 datapoints, Adjacent bins only)
log(ΣSFR) [M☉ yr⁻¹ kpc⁻²]
-4.57 to -2.92
-2.92 to -2.59
-2.59 to -2.34
-2.34 to -2.08
-2.08 to -1.77
-1.77 to 0.02
HII rMZR scatter (D16)
HII rMZR median trends (D16)
HII rMZR scatter (PG16)
HII rMZR median trends (PG16)
HII rMZR scatter (O3N2-M13)
HII rMZR median trends (O3N2-M13)
HII rMZR scatter (O3N2-C20)
HII rMZR median trends (O3N2-C20)
[O/H] (D16)
[O/H] (PG16)
[O/H] (O3N2-M13)
[O/H] (O3N2-C20)
log Σ* (M☉ kpc⁻²)

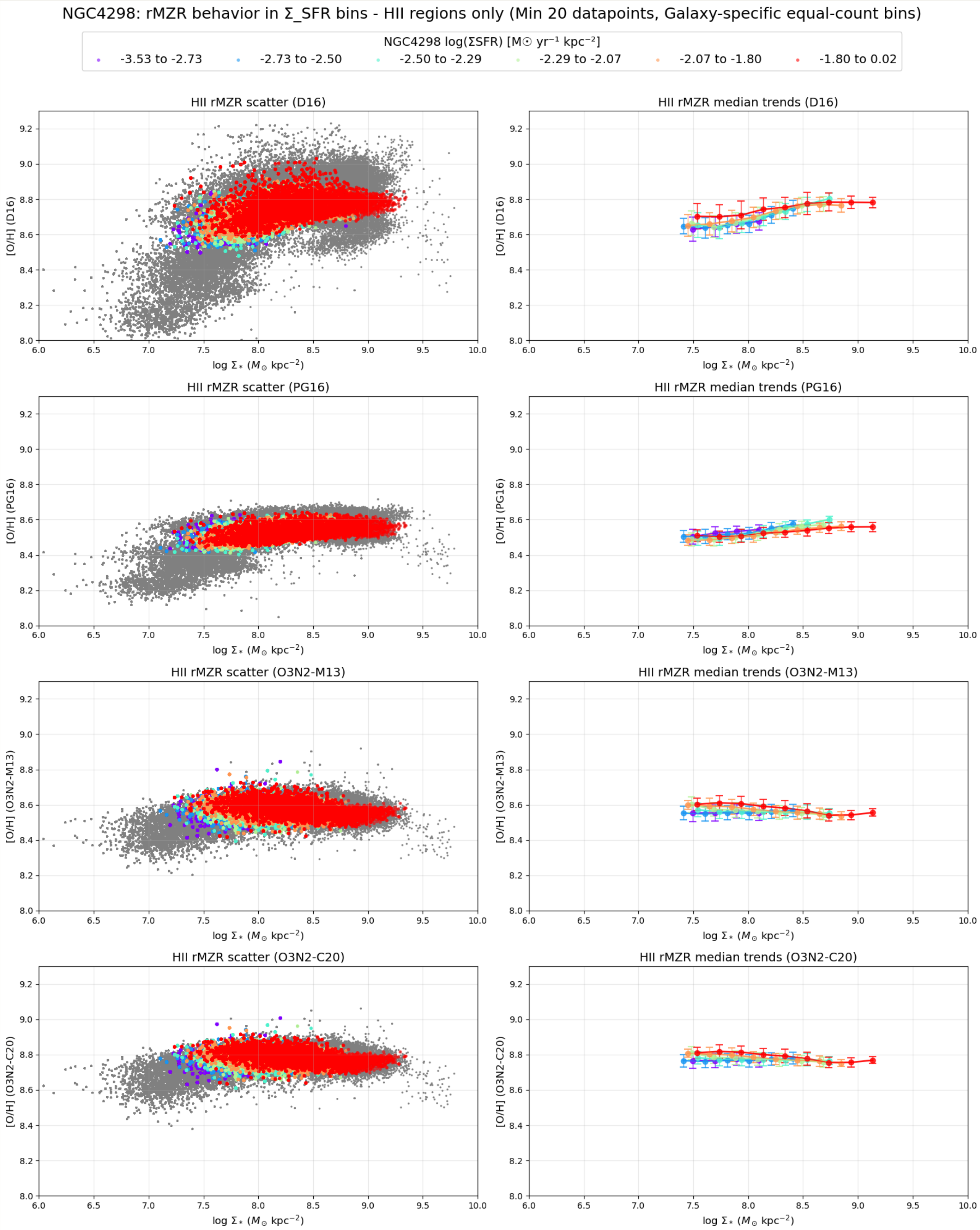
Individual

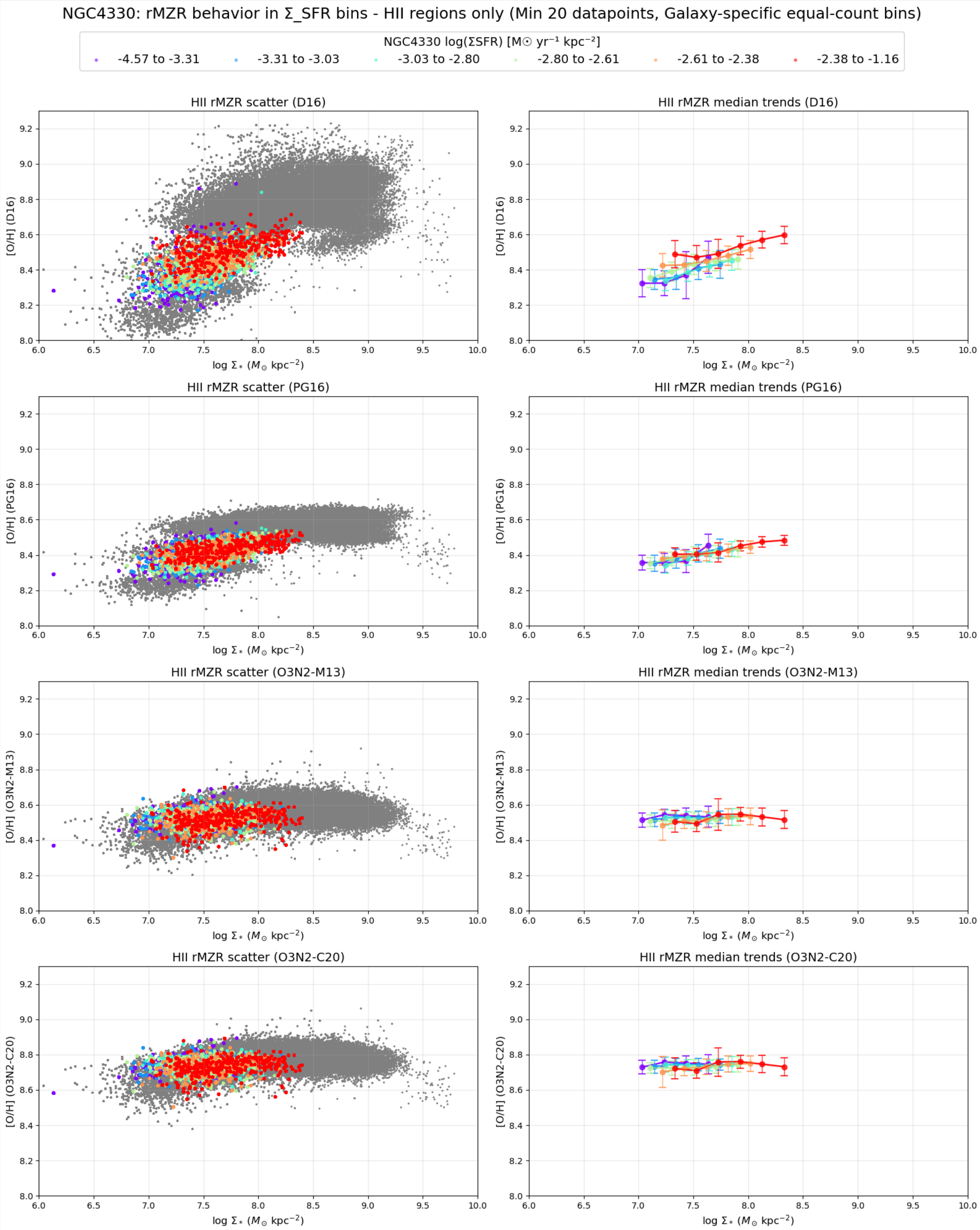

IC3392: rMZR behavior in Σ_SFR bins - HII regions only (Min 20 datapoints, Galaxy-specific equal-count bins)
IC3392 log(ΣSFR) [M☉ yr⁻¹ kpc⁻²]
-4.09 to -2.24
-2.24 to -2.02
-2.02 to -1.86
-1.86 to -1.73
-1.73 to -1.56
-1.56 to -0.90
HII rMZR scatter (D16)
HII rMZR median trends (D16)
[O/H] (D16)
HII rMZR scatter (PG16)
HII rMZR median trends (PG16)
[O/H] (PG16)
HII rMZR scatter (O3N2-M13)
HII rMZR median trends (O3N2-M13)
[O/H] (O3N2-M13)
HII rMZR scatter (O3N2-C20)
HII rMZR median trends (O3N2-C20)
[O/H] (O3N2-C20)
log Σ* (M☉ kpc⁻²)

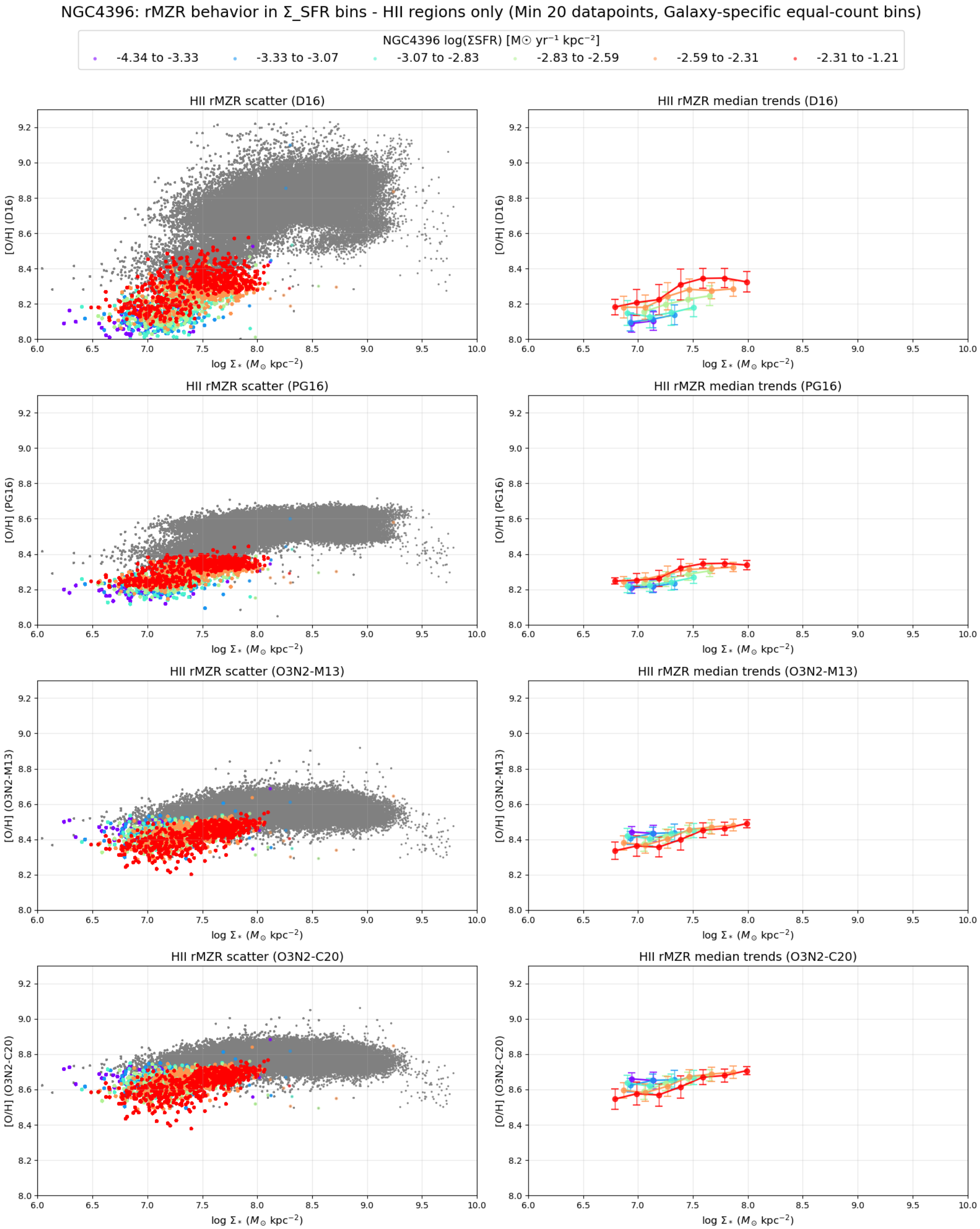

NGC4064: rMZR behavior in Σ_SFR bins - HII regions only (Min 20 datapoints, Galaxy-specific equal-count bins)
NGC4064 log(ΣSFR) [M☉ yr⁻¹ kpc⁻²]
-3.99 to -2.39
-2.39 to -2.05
-2.05 to -1.68
-1.68 to -1.28
-1.28 to -0.79
-0.79 to -0.05
HII rMZR scatter (D16)
HII rMZR median trends (D16)
[O/H] (D16)
HII rMZR scatter (PG16)
HII rMZR median trends (PG16)
[O/H] (PG16)
HII rMZR scatter (O3N2-M13)
HII rMZR median trends (O3N2-M13)
[O/H] (O3N2-M13)
HII rMZR scatter (O3N2-C20)
HII rMZR median trends (O3N2-C20)
[O/H] (O3N2-C20)
log Σ* (M☉ kpc⁻²)


NGC4192: rMZR behavior in Σ_SFR bins - HII regions only (Min 20 datapoints, Galaxy-specific equal-count bins)
NGC4192 log(ΣSFR) [M☉ yr⁻¹ kpc⁻²]
-3.96 to -3.03
-3.03 to -2.78
-2.78 to -2.56
-2.56 to -2.34
-2.34 to -2.04
-2.04 to -0.07
HII rMZR scatter (D16)
HII rMZR median trends (D16)
HII rMZR scatter (PG16)
HII rMZR median trends (PG16)
HII rMZR scatter (O3N2-M13)
HII rMZR median trends (O3N2-M13)
HII rMZR scatter (O3N2-C20)
HII rMZR median trends (O3N2-C20)
[O/H] (D16)
[O/H] (PG16)
[O/H] (O3N2-M13)
[O/H] (O3N2-C20)
log Σ* (M⊙ kpc⁻²)

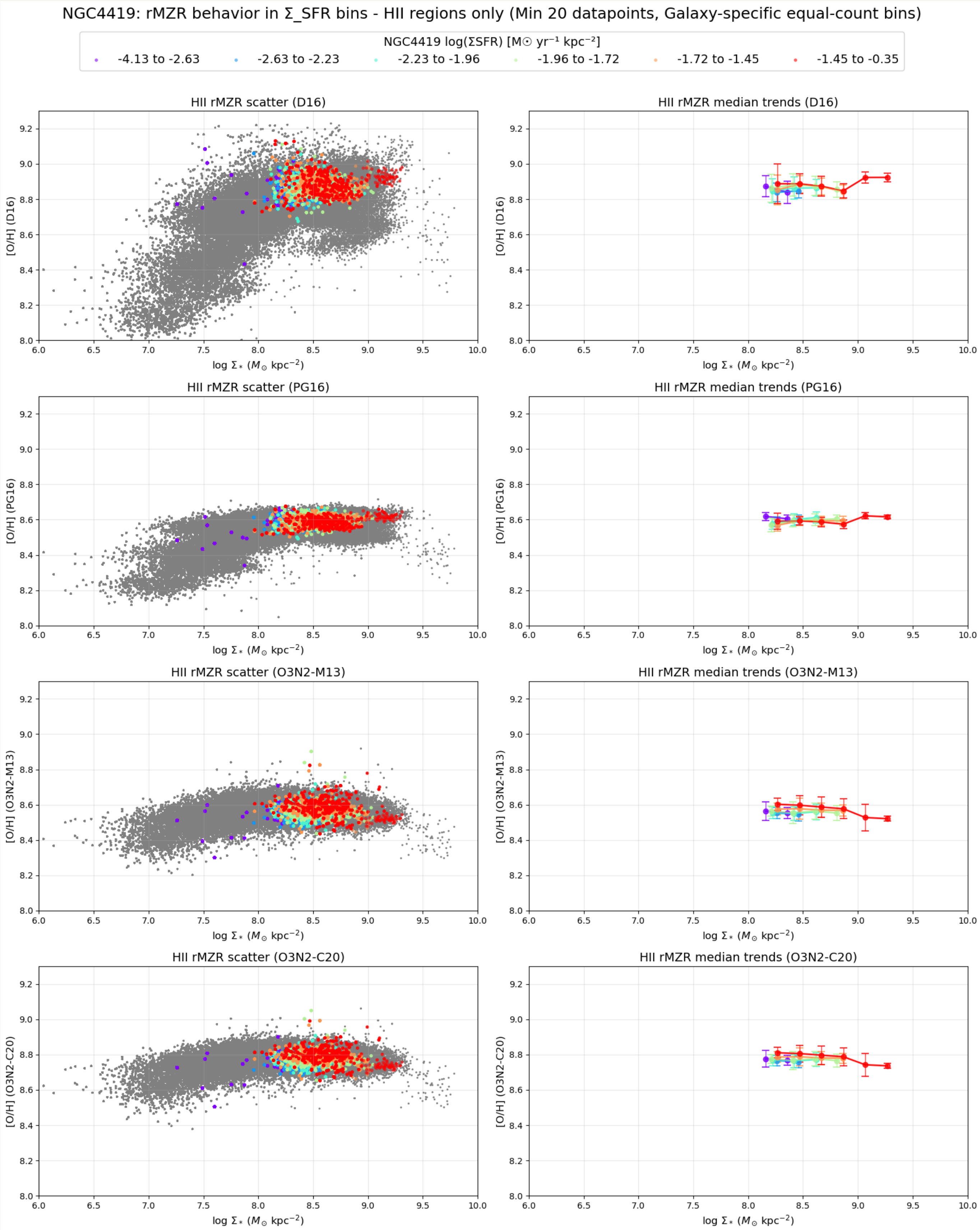

NGC4293: rMZR behavior in Σ_SFR bins - HII regions only (Min 20 datapoints, Galaxy-specific equal-count bins)
NGC4293 log(ΣSFR) [M☉ yr⁻¹ kpc⁻²]
-4.16 to -3.84
-3.84 to -3.67
-3.67 to -3.52
-3.52 to -2.97
-2.97 to -2.43
-2.43 to -0.94
HII rMZR scatter (D16)
HII rMZR median trends (D16)
HII rMZR scatter (PG16)
HII rMZR median trends (PG16)
HII rMZR scatter (O3N2-M13)
HII rMZR median trends (O3N2-M13)
HII rMZR scatter (O3N2-C20)
HII rMZR median trends (O3N2-C20)
[O/H] (D16)
[O/H] (PG16)
[O/H] (O3N2-M13)
[O/H] (O3N2-C20)
log Σ* ($M_\odot$ kpc^{-2})

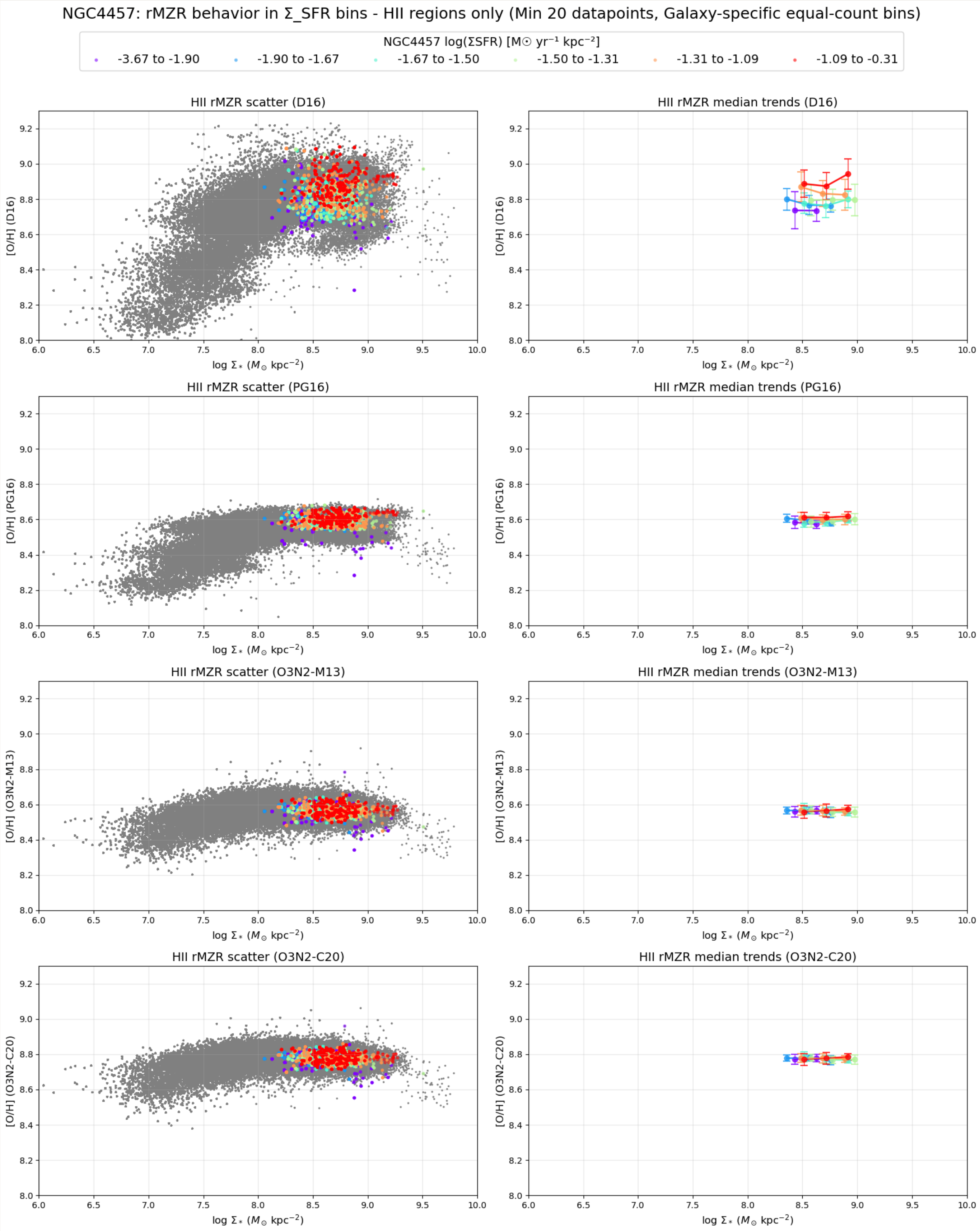

NGC4298: rMZR behavior in Σ_SFR bins - HII regions only (Min 20 datapoints, Galaxy-specific equal-count bins)
NGC4298 log(ΣSFR) [M☉ yr⁻¹ kpc⁻²]
-3.53 to -2.73
-2.73 to -2.50
-2.50 to -2.29
-2.29 to -2.07
-2.07 to -1.80
-1.80 to 0.02
HII rMZR scatter (D16)
HII rMZR median trends (D16)
HII rMZR scatter (PG16)
HII rMZR median trends (PG16)
HII rMZR scatter (O3N2-M13)
HII rMZR median trends (O3N2-M13)
HII rMZR scatter (O3N2-C20)
HII rMZR median trends (O3N2-C20)
[O/H] (D16)
[O/H] (PG16)
[O/H] (O3N2-M13)
[O/H] (O3N2-C20)
log Σ* ($M_\odot$ kpc^{-2})

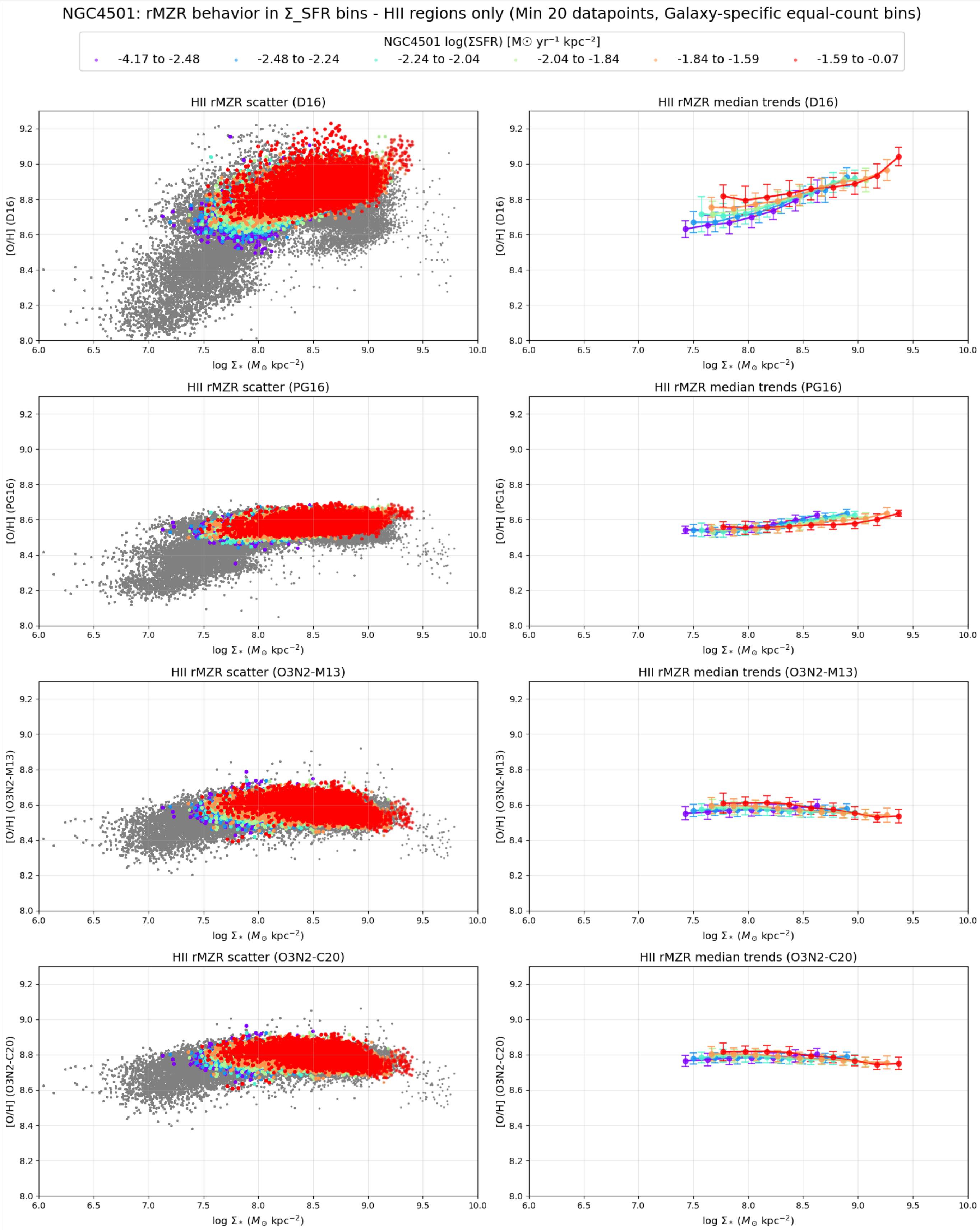

NGC4330: rMZR behavior in Σ_SFR bins - HII regions only (Min 20 datapoints, Galaxy-specific equal-count bins)
NGC4330 log(ΣSFR) [M☉ yr⁻¹ kpc⁻²]
-4.57 to -3.31
-3.31 to -3.03
-3.03 to -2.80
-2.80 to -2.61
-2.61 to -2.38
-2.38 to -1.16
HII rMZR scatter (D16)
HII rMZR median trends (D16)
[O/H] (D16)
HII rMZR scatter (PG16)
HII rMZR median trends (PG16)
[O/H] (PG16)
HII rMZR scatter (O3N2-M13)
HII rMZR median trends (O3N2-M13)
[O/H] (O3N2-M13)
HII rMZR scatter (O3N2-C20)
HII rMZR median trends (O3N2-C20)
[O/H] (O3N2-C20)
log Σ* (M☉ kpc⁻²)

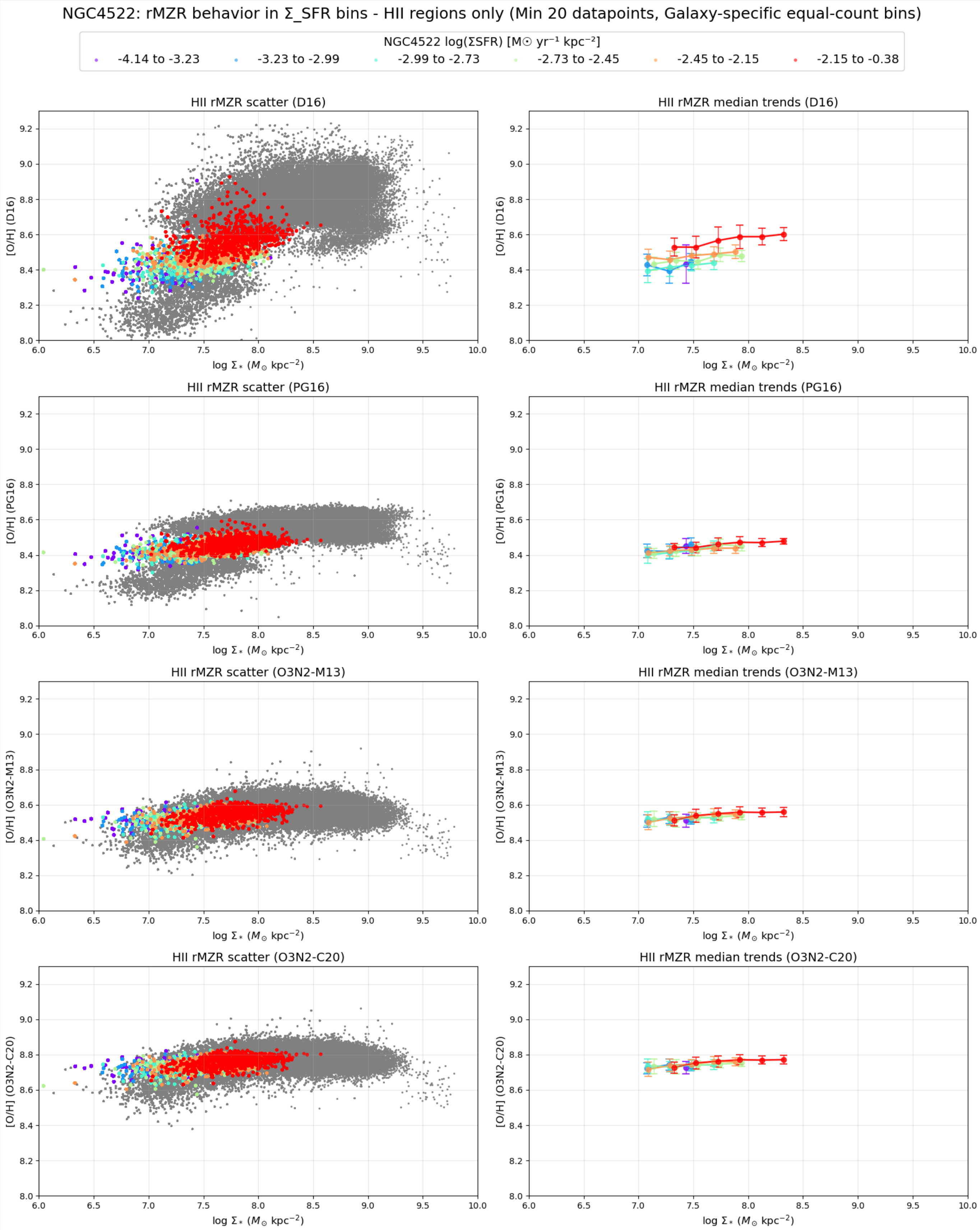


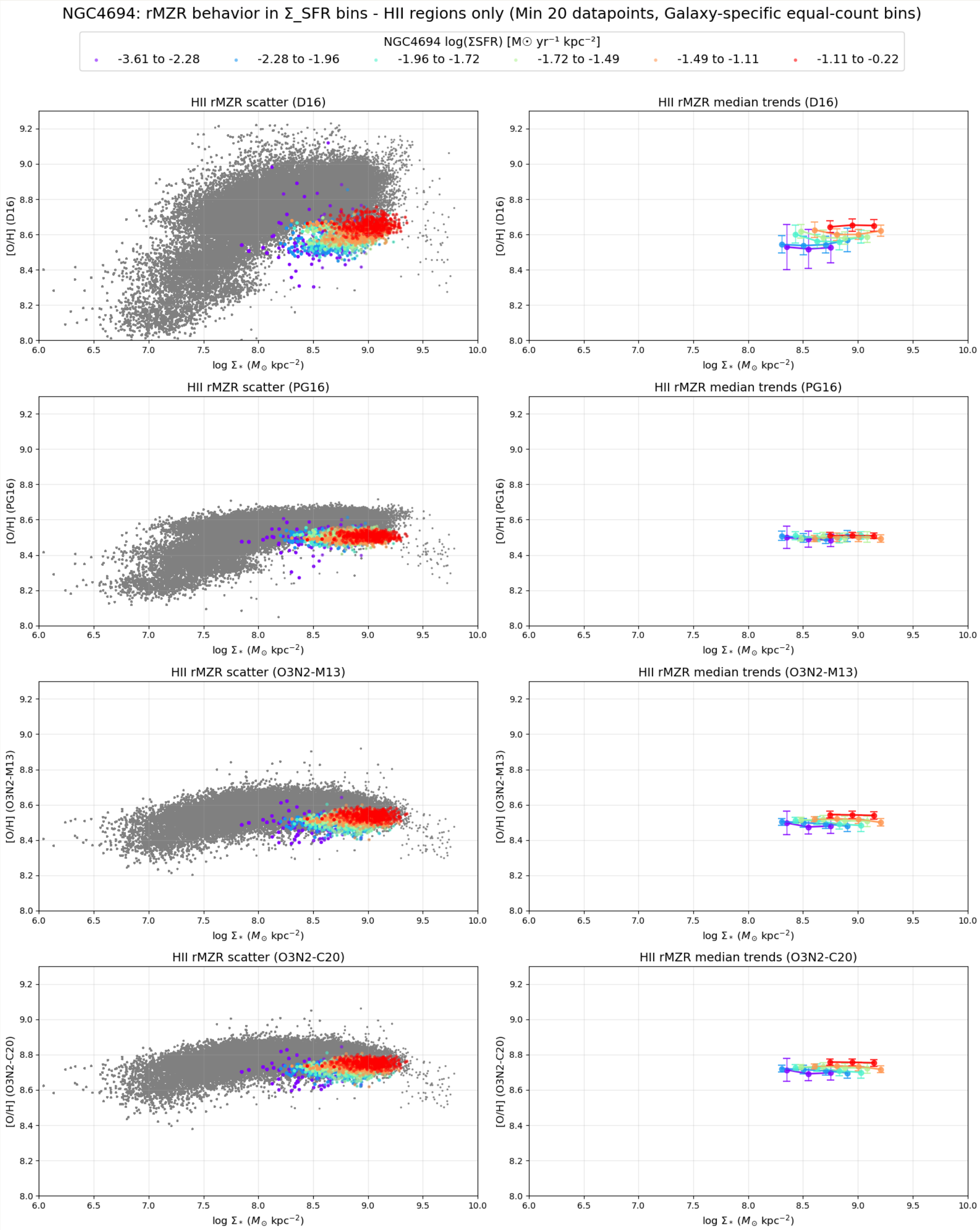
NGC4396: rMZR behavior in Σ_SFR bins - HII regions only (Min 20 datapoints, Galaxy-specific equal-count bins)
NGC4396 log(ΣSFR) [M☉ yr⁻¹ kpc⁻²]
-4.34 to -3.33
-3.33 to -3.07
-3.07 to -2.83
-2.83 to -2.59
-2.59 to -2.31
-2.31 to -1.21
HII rMZR scatter (D16)
HII rMZR median trends (D16)
HII rMZR scatter (PG16)
HII rMZR median trends (PG16)
HII rMZR scatter (O3N2-M13)
HII rMZR median trends (O3N2-M13)
HII rMZR scatter (O3N2-C20)
HII rMZR median trends (O3N2-C20)
[O/H] (D16)
[O/H] (PG16)
[O/H] (O3N2-M13)
[O/H] (O3N2-C20)
log Σ* ($M_\odot$ kpc^{-2})

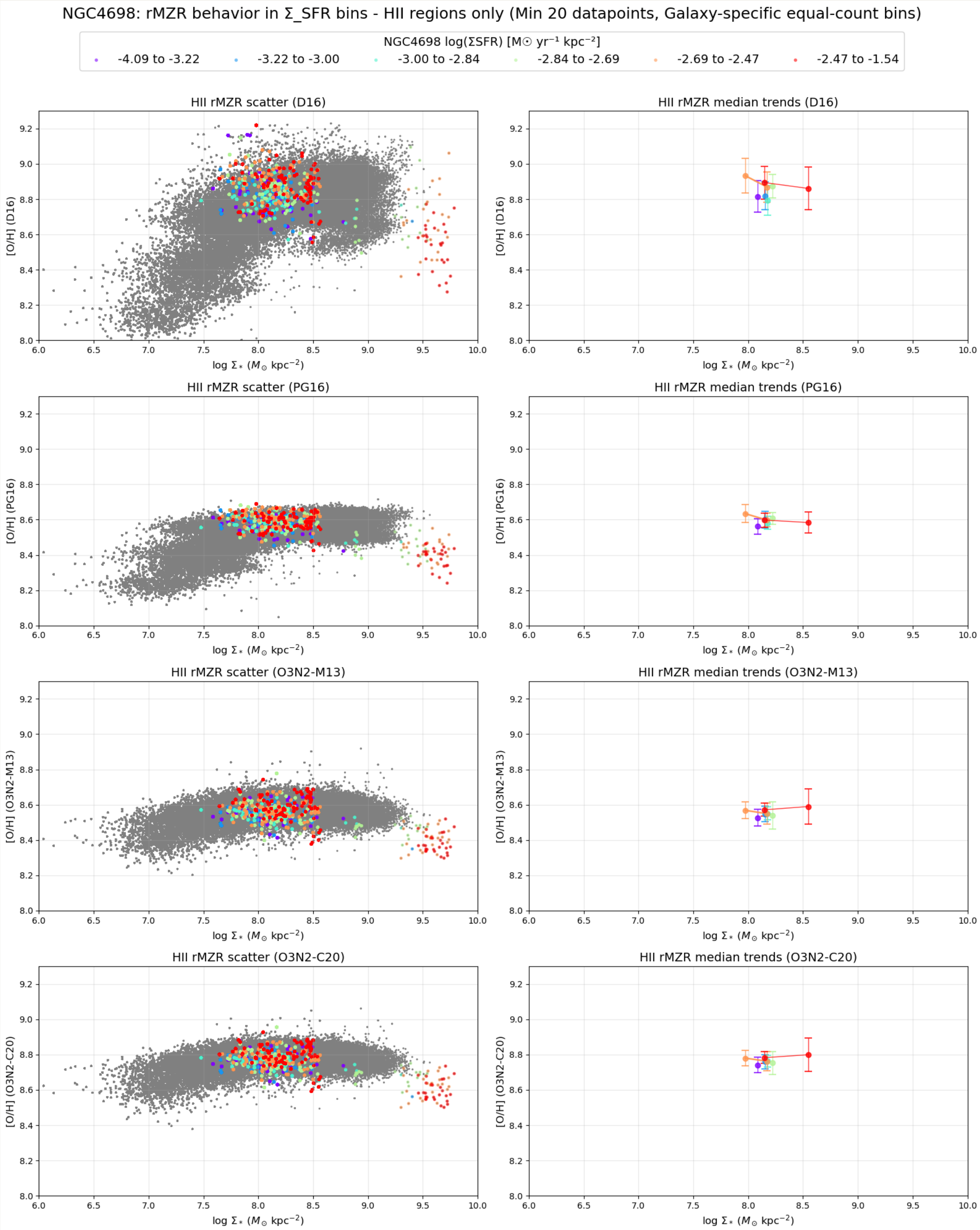

NGC4419: rMZR behavior in Σ_SFR bins - HII regions only (Min 20 datapoints, Galaxy-specific equal-count bins)
NGC4419 log(ΣSFR) [M☉ yr⁻¹ kpc⁻²]
-4.13 to -2.63
-2.63 to -2.23
-2.23 to -1.96
-1.96 to -1.72
-1.72 to -1.45
-1.45 to -0.35
HII rMZR scatter (D16)
HII rMZR median trends (D16)
[O/H] (D16)
HII rMZR scatter (PG16)
HII rMZR median trends (PG16)
[O/H] (PG16)
HII rMZR scatter (O3N2-M13)
HII rMZR median trends (O3N2-M13)
[O/H] (O3N2-M13)
HII rMZR scatter (O3N2-C20)
HII rMZR median trends (O3N2-C20)
[O/H] (O3N2-C20)
log Σ* ($M_\odot$ kpc^{-2})


NGC4457: rMZR behavior in Σ_SFR bins - HII regions only (Min 20 datapoints, Galaxy-specific equal-count bins)
NGC4457 log(ΣSFR) [M☉ yr⁻¹ kpc⁻²]
-3.67 to -1.90
-1.90 to -1.67
-1.67 to -1.50
-1.50 to -1.31
-1.31 to -1.09
-1.09 to -0.31
HII rMZR scatter (D16)
HII rMZR median trends (D16)
HII rMZR scatter (PG16)
HII rMZR median trends (PG16)
HII rMZR scatter (O3N2-M13)
HII rMZR median trends (O3N2-M13)
HII rMZR scatter (O3N2-C20)
HII rMZR median trends (O3N2-C20)
[O/H] (D16)
[O/H] (PG16)
[O/H] (O3N2-M13)
[O/H] (O3N2-C20)
log Σ* (M☉ kpc⁻²)


NGC4501: rMZR behavior in Σ_SFR bins - HII regions only (Min 20 datapoints, Galaxy-specific equal-count bins)
NGC4501 log(ΣSFR) [M☉ yr⁻¹ kpc⁻²]
-4.17 to -2.48
-2.48 to -2.24
-2.24 to -2.04
-2.04 to -1.84
-1.84 to -1.59
-1.59 to -0.07
HII rMZR scatter (D16)
HII rMZR median trends (D16)
HII rMZR scatter (PG16)
HII rMZR median trends (PG16)
HII rMZR scatter (O3N2-M13)
HII rMZR median trends (O3N2-M13)
HII rMZR scatter (O3N2-C20)
HII rMZR median trends (O3N2-C20)
[O/H] (D16)
[O/H] (PG16)
[O/H] (O3N2-M13)
[O/H] (O3N2-C20)
log Σ_* ($M_\odot$ kpc^{-2})


NGC4522: rMZR behavior in Σ_SFR bins - HII regions only (Min 20 datapoints, Galaxy-specific equal-count bins)
NGC4522 log(ΣSFR) [M☉ yr⁻¹ kpc⁻²]
-4.14 to -3.23
-3.23 to -2.99
-2.99 to -2.73
-2.73 to -2.45
-2.45 to -2.15
-2.15 to -0.38
HII rMZR scatter (D16)
HII rMZR median trends (D16)
[O/H] (D16)
HII rMZR scatter (PG16)
HII rMZR median trends (PG16)
[O/H] (PG16)
HII rMZR scatter (O3N2-M13)
HII rMZR median trends (O3N2-M13)
[O/H] (O3N2-M13)
HII rMZR scatter (O3N2-C20)
HII rMZR median trends (O3N2-C20)
[O/H] (O3N2-C20)
log Σ_* ($M_\odot$ kpc^{-2})

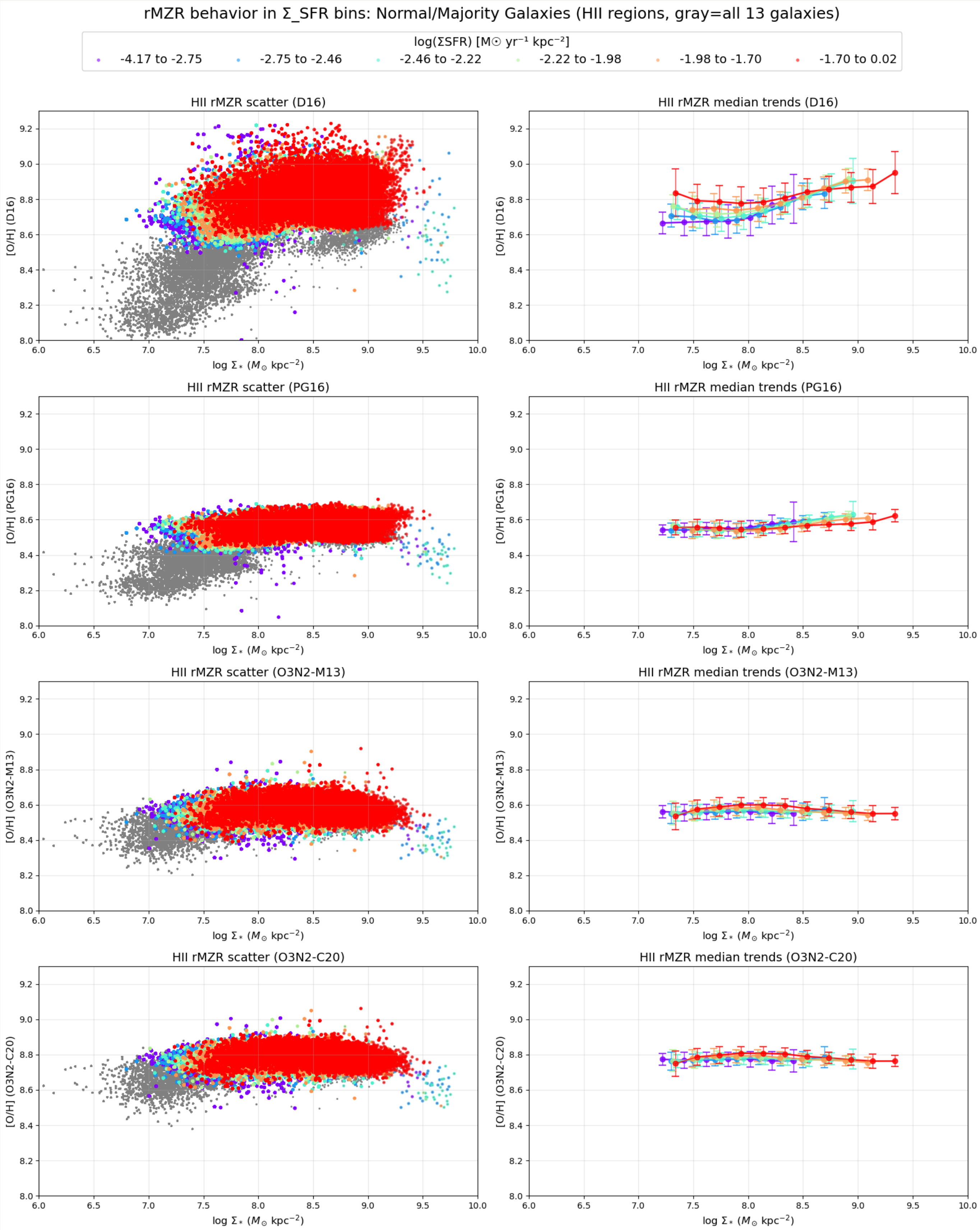

NGC4694: rMZR behavior in Σ_SFR bins - HII regions only (Min 20 datapoints, Galaxy-specific equal-count bins)
NGC4694 log(ΣSFR) [M☉ yr⁻¹ kpc⁻²]
-3.61 to -2.28
-2.28 to -1.96
-1.96 to -1.72
-1.72 to -1.49
-1.49 to -1.11
-1.11 to -0.22
HII rMZR scatter (D16)
HII rMZR median trends (D16)
HII rMZR scatter (PG16)
HII rMZR median trends (PG16)
HII rMZR scatter (O3N2-M13)
HII rMZR median trends (O3N2-M13)
HII rMZR scatter (O3N2-C20)
HII rMZR median trends (O3N2-C20)
[O/H] (D16)
[O/H] (PG16)
[O/H] (O3N2-M13)
[O/H] (O3N2-C20)
log Σ* (M☉ kpc⁻²)

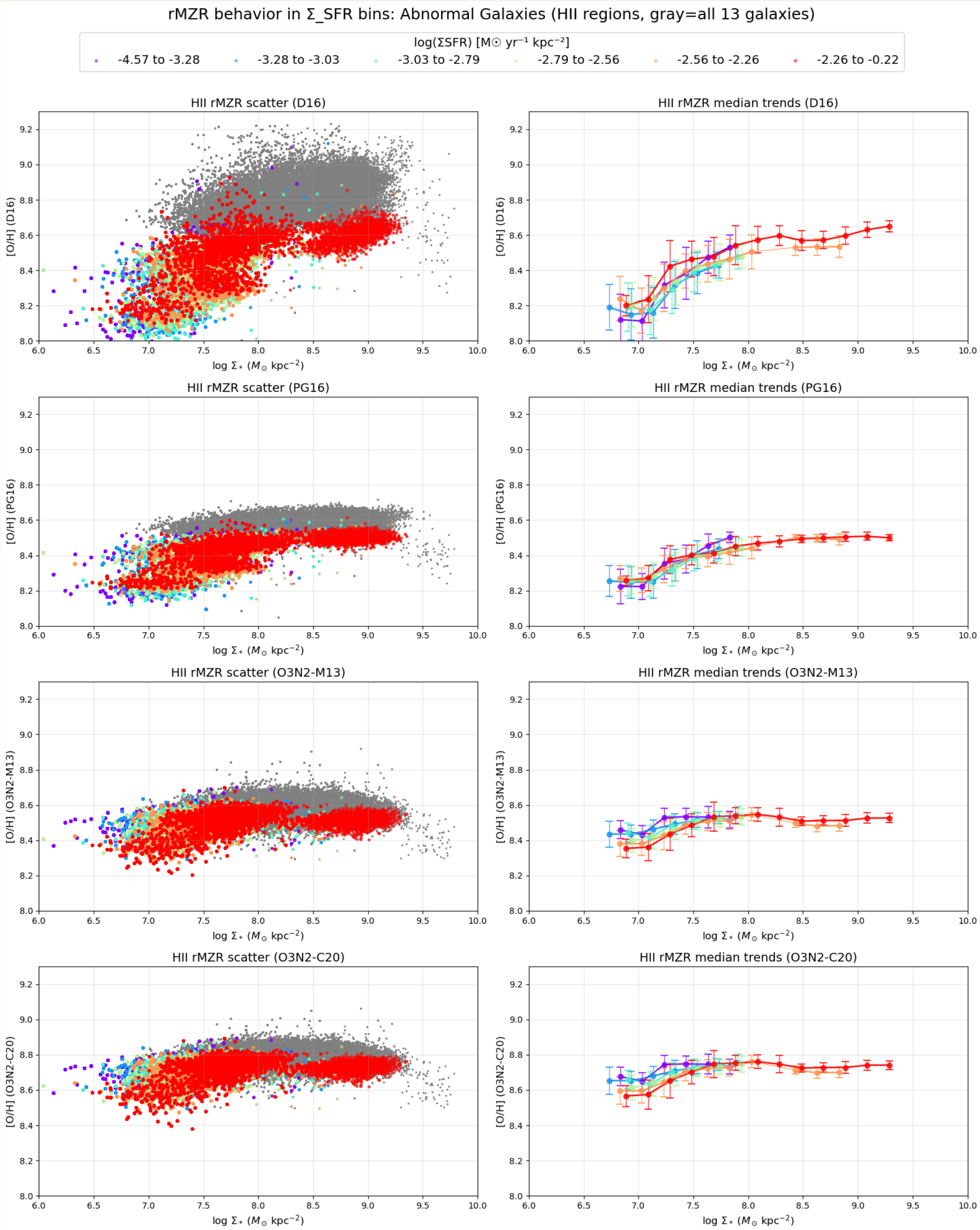


What if we try seperate the combinded rMZR plots into two branches: outliers and majority

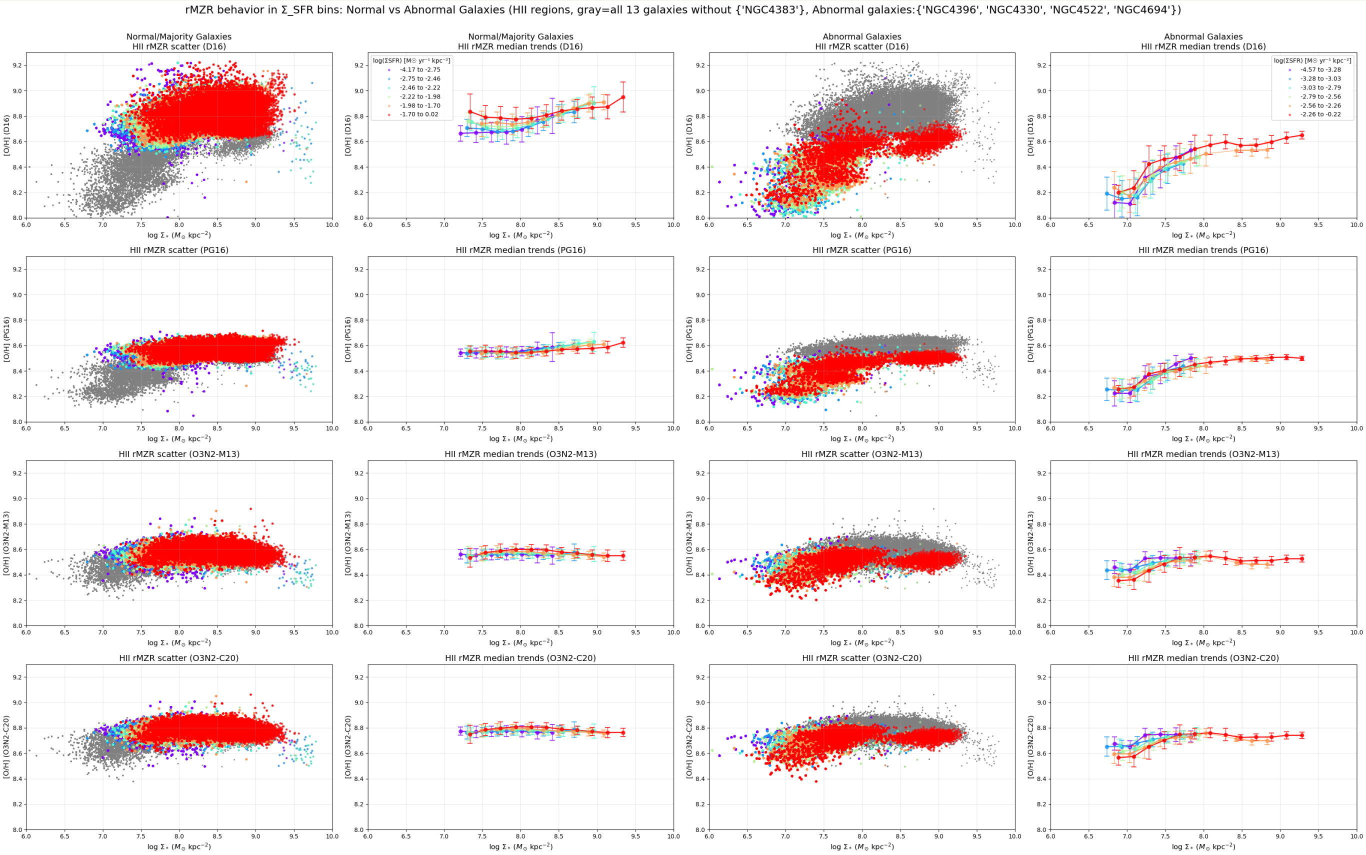
Remember, except the NGC4383 that has been excluded, the other abnormal galaxies are: NGC4330, NGC4396, NGC4522 and NGC4694.

1. NGC4330: low mass end, metallicity anti-correlate with SFR, match expected trend
2. NGC4396: low mass end, metallicity anti-correlate with SFR, match expected trend
3. NGC4522: low mass end, can't see clear trend
4. NGC4694: high mass end, metallicity anti-correlate with SFR, match expected trend


rMZR behavior in Σ_SFR bins: Normal/Majority Galaxies (HII regions, gray=all 13 galaxies)
log(ΣSFR) [M☉ yr⁻¹ kpc⁻²]
-4.17 to -2.75
-2.75 to -2.46
-2.46 to -2.22
-2.22 to -1.98
-1.98 to -1.70
-1.70 to 0.02
HII rMZR scatter (D16)
HII rMZR median trends (D16)
[O/H] (D16)
HII rMZR scatter (PG16)
HII rMZR median trends (PG16)
[O/H] (PG16)
HII rMZR scatter (O3N2-M13)
HII rMZR median trends (O3N2-M13)
[O/H] (O3N2-M13)
HII rMZR scatter (O3N2-C20)
HII rMZR median trends (O3N2-C20)
[O/H] (O3N2-C20)
log Σ* ($M_\odot$ kpc^{-2})



So now we can may conclude that the anti-correlation between Z and Σ_{SFR} in low mass end are driven by 3 outliers: NGC4330, NGC4396, NGC4522. Below I combine two above plots together


rMZR behavior in Σ_SFR bins: Normal vs Abnormal Galaxies (HII regions, gray=all 13 galaxies without {'NGC4383'}, Abnormal galaxies:{'NGC4396', 'NGC4330', 'NGC4522', 'NGC4694'})
Normal/Majority Galaxies
HII rMZR scatter (D16)
Normal/Majority Galaxies
HII rMZR median trends (D16)
log(ΣSFR) [M☉ yr⁻¹ kpc⁻²]
-4.17 to -2.75
-2.75 to -2.46
-2.46 to -2.22
-2.22 to -1.98
-1.98 to -1.70
-1.70 to 0.02
Abnormal Galaxies
HII rMZR scatter (D16)
Abnormal Galaxies
HII rMZR median trends (D16)
log(ΣSFR) [M☉ yr⁻¹ kpc⁻²]
-4.57 to -3.28
-3.28 to -3.03
-3.03 to -2.79
-2.79 to -2.56
-2.56 to -2.26
-2.26 to -0.22
[O/H] (D16)
HII rMZR scatter (PG16)
HII rMZR median trends (PG16)
[O/H] (PG16)
HII rMZR scatter (O3N2-M13)
HII rMZR median trends (O3N2-M13)
[O/H] (O3N2-M13)
HII rMZR scatter (O3N2-C20)
HII rMZR median trends (O3N2-C20)
[O/H] (O3N2-C20)
log Σ* ($M_\odot$ kpc^{-2})